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NAAC
GRADE A+
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

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Github link : https://github.com/AarushCS777/Aarush_BusinessAnalytics

1. INTRODUCTION

Power BI is a business intelligence and data visualization tool developed by Microsoft. It is mainly used to analyse data and present the results in the form of charts, graphs, tables and dashboards. It helps users understand large amounts of data more easily.

In this project, I used Power BI to analyse the **Sample Superstore dataset**. The dataset contains information about orders, customers, products, sales, profit and regions.

The main purpose of this project is to understand the sales performance of different products, categories and regions. I created a dashboard using different Power BI visualizations. The dashboard gives a quick overview of the total sales, total profit, quantity sold and number of orders.

I also created charts to compare sales and profit between different categories and regions. A Top 10 Products chart was also created to identify the products having higher sales.

This project helped me understand how raw data can be converted into useful information using data visualization.

2. OBJECTIVES

The main objectives of this project are:

- To understand the basic use of Microsoft Power BI.
- To work with the Sample Superstore dataset.
- To analyse sales and profit data.
- To create different types of visualizations.
- To compare sales between different categories.
- To compare sales and profit between different regions.
- To identify the top products based on sales.
- To create an interactive dashboard using slicers.
- To understand important business patterns from the data.

3. DATASET USED

For this project, I used the **Sample Superstore dataset** from Kaggle.

The dataset contains information about sales transactions. It includes details about customers, products, orders, shipping and regions.

Some of the important columns used in this project are:

Column	Description
Order ID	Identifies an order

Column	Description
Order Date	Date on which the order was placed
Ship Date	Date on which the order was shipped
Customer ID	Identifies the customer
Customer Name	Name of the customer
Region	Region of the order
Category	Main product category
Sub-Category	Type of product
Product Name	Name of the product
Sales	Sales amount
Quantity	Number of products sold
Discount	Discount given
Profit	Profit generated

The main columns used for creating the dashboard were **Sales, Profit, Quantity, Order ID, Category, Sub-Category, Region and Product Name.**

Before creating the visualizations, the data types of important columns were checked in Power Query. The Order Date and Ship Date columns were converted into date format, while numerical fields such as Sales, Quantity and Profit were set to suitable numerical data types.

4. TOOL USED

Microsoft Power BI Desktop

Microsoft Power BI Desktop was used for this project.

The main features used were:

- Power Query for checking and preparing the data.
- Data pane for selecting the required fields.
- Card visuals for displaying important values.
- Column charts for comparing categories and regions.
- Bar charts for comparing products and sub-categories.
- Slicers for filtering the dashboard.

Power BI was useful because the data could be represented visually instead of looking at the values in a table.

5. DASHBOARD CREATION

After loading the Sample Superstore dataset into Power BI, I selected the fields required for the

analysis.

First, I created four cards to show the main values:

- Total Sales
- Total Profit
- Total Quantity
- Total Orders

After creating the cards, different charts were added to analyse sales and profit.

The dashboard contains the following visualizations:

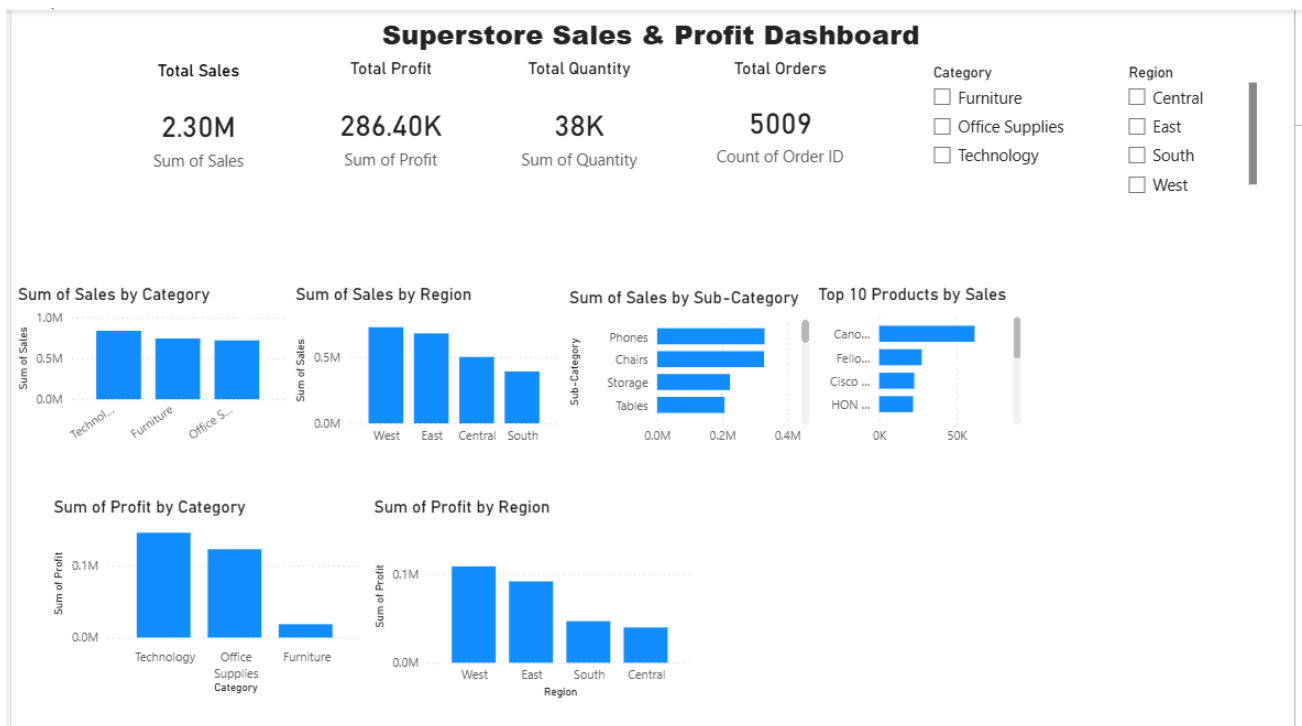
1. Sales by Category
2. Sales by Region
3. Sales by Sub-Category
4. Top 10 Products by Sales
5. Profit by Category
6. Profit by Region

I also added two slicers:

- Category
- Region

These slicers allow the dashboard to be filtered based on a particular category or region.

6. DASHBOARD SCREENSHOT



7. VISUALIZATIONS AND ANALYSIS

7.1 Total Sales

The Total Sales card shows the total sales value in the dataset.

The dashboard shows approximately:

Total Sales = 2.30M

This gives a quick idea about the overall sales generated from the orders in the dataset.

7.2 Total Profit

The Total Profit card shows the total profit generated from the sales.

The dashboard shows approximately:

Total Profit = 286.40K

This value is useful for understanding the overall profitability of the business.

7.3 Total Quantity

The Total Quantity card shows the total number of products sold.

The dashboard shows approximately:

Total Quantity = 38K

This helps to understand the total number of units sold through the orders in the dataset.

7.4 Total Orders

The Total Orders card shows the number of unique orders.

The dashboard shows:

Total Orders = 5009

This gives an idea of the number of different orders present in the dataset.

8. SALES BY CATEGORY

The **Sales by Category** chart is used to compare sales between the three main categories.

The categories are:

- Technology
- Furniture
- Office Supplies

From the chart, **Technology has the highest sales** among the three categories. Furniture is next, followed by Office Supplies.

This shows that Technology products are an important source of sales in the dataset.

The chart makes it easier to compare the categories instead of checking the sales values manually.

9. SALES BY REGION

The **Sales by Region** chart compares the sales generated from different regions.

The four regions are:

- West
- East
- Central
- South

From the visualization, the **West region has the highest sales**. The East region is also performing well, while Central and South have comparatively lower sales.

This visualization helps management understand which regions are contributing more to total sales.

10. SALES BY SUB-CATEGORY

The **Sales by Sub-Category** chart provides more detailed information about product sales.

Unlike Category, which gives a broad classification, Sub-Category provides information about individual types of products.

From the chart, sub-categories such as **Phones and Chairs** have relatively high sales.

This visualization can help identify the types of products that are selling more and can be useful when studying product performance.

11. TOP 10 PRODUCTS BY SALES

The **Top 10 Products by Sales** chart shows the products with the highest sales values.

This visualization was created using Product Name and Sales. The products were arranged from higher sales to lower sales.

The chart helps to identify the products that contribute more to the total sales.

The highest product in the chart has sales of more than **50K**.

This type of analysis can be useful for understanding which individual products are performing well.

12. PROFIT BY CATEGORY

The **Profit by Category** chart compares the profit generated by the three categories.

From the visualization, **Technology has the highest profit**.

Office Supplies also contributes a good amount of profit, while Furniture has considerably lower profit compared to Technology.

This shows that looking only at sales is not enough. Profit should also be considered when analysing the performance of a category.

13. PROFIT BY REGION

The **Profit by Region** chart compares profit between the four regions.

The **West region has the highest profit**, followed by the East region.

Central and South have lower profit compared to West and East.

This chart helps to understand which regions are generating better returns for the business.

14. SLICERS

Two slicers were added to make the dashboard interactive.

Category Slicer

The Category slicer allows the user to select:

- Furniture
- Office Supplies
- Technology

When a category is selected, the other visuals can be viewed based on that selection.

Region Slicer

The Region slicer allows the user to select:

- Central
- East
- South
- West

This helps the user analyse the dashboard for a particular region.

The slicers make the dashboard easier to explore and understand.

15. KEY FINDINGS

After analysing the dashboard, some important findings were identified.

1. The total sales of the dataset are approximately **2.30M**.
2. The total profit is approximately **286.40K**.
3. Around **38K products** were sold according to the Quantity field.
4. The dataset contains **5009 unique orders**.
5. **Technology** is the category with the highest sales.
6. Technology also has the highest profit among the three categories.
7. The **West region** has the highest sales.
8. The West region also has the highest profit.
9. **Phones and Chairs** are among the higher-selling sub-categories.
10. The Top 10 Products chart helps identify the individual products that contribute more to sales.

These findings provide a simple overview of the business performance in the Sample Superstore dataset

